Tech Domain Platform — Checklist Test Report (Batch 1: first 10%)

Website: https://dev.techdomain.com.au · API: https://api.dev.techdomain.com.au/api · Date: 2026-10-14 · Tool: Playwright (API + browser) · Tests: tests/checklist/*.spec.js · Evidence: evidence/Checklist/

1. In short

- We tested the **first 455 items (10%)** of the 4550-item functionality checklist: notifications, emails/sockets rules that are visible from the API, chat & inbox rules, roles & permissions, login / forgot / reset password, OTP, sign-up forms and the "Complete Your Profile" wizard.
- For every feature we ran **normal cases, wrong-input cases, unusual situations and limit values** (for example password length 7 / 8 / 20 / 21, list sizes 0 / 1 / max / max+1, ids like 0, -1, "abc").
- **473 checks ran: 433 passed, 40 failed (92% pass rate).**
- **21 high-severity findings** need developer attention first (server crashes on bad input, validation only in the browser, security notes). Full list in section 3.
- A live dashboard is in the project at **reports/checklist-dashboard.html**; it is rebuilt after every batch, so progress keeps updating as more of the checklist is tested.

2. Results per area

Area	Checks	Passed	Failed	Pass rate
Authentication API	103	91	12	88%
Authentication screens (UI)	89	89	0	100%
Notifications API (§1.3)	66	52	14	79%
Chat & inbox rules (§4)	67	61	6	91%
Roles & permissions (§6)	116	113	3	97%
Onboarding — Complete Your Profile wizard	32	27	5	84%
Total	**473**	**433**	**40**	**92%**

By kind of test

Kind	Meaning	Checks	Passed	Failed
Positive	Normal use (should work)	156	156	0
Negative	Wrong input (should be refused)	178	162	16
Edge case	Unusual situation	52	47	5
Boundary value	Limit value (min / max / just outside)	87	68	19

3. Findings (things to fix or decide)

High = server error, security or data problem · Medium = behaves differently from the checklist · Low = informational, product decision

Severity	ID / area	Check	Should happen	Did happen	Note
High	LG-N13 01-AuthAPI	Login rejected: email as number	HTTP 400/401/404, no token, generic message	HTTP 500 token=false msg="Internal server error"	
High	SU-N01 01-AuthAPI	Signup again with same unverified email → 403 (go to OTP)	HTTP 403	HTTP 500 msg="User already exists with this email, OTP code sent"	
High	OTP-N06 01-AuthAPI	OTP verify rejected: code as number	HTTP 400/401/404, no token	HTTP 500 token=false msg="Something went wrong"	
High	SU-N02 01-AuthAPI	Signup rejected: existing verified email	HTTP 400/403/409	HTTP 500 msg="User already exists."	
High	RT-E01 01-AuthAPI	Re-using an already-exchanged refresh token	HTTP 401/403 (rotated) — informational if accepted	HTTP 201	OLD REFRESH TOKEN STILL ACCEPTED (no rotation) — security note
High	NB-B-others-0 03-NotificationsAPI	GET /notifications/others limit=0	HTTP 200/400; items ≤ limit	HTTP 500 items=0 (undefined notifications)	SERVER ERROR 500
High	NB-B-others--1 03-NotificationsAPI	GET /notifications/others limit=-1	HTTP 200/400; items ≤ limit	HTTP 500 items=0 (undefined notifications)	SERVER ERROR 500
High	NR-N01 03-NotificationsAPI	Marking ANOTHER user's notification read is refused with 404/403 (not 500)	HTTP 403/404	HTTP 500 msg="Internal server error"	CONFIRMED BUG: 500 Internal Server Error
High	NR-B01 03-NotificationsAPI	PATCH /notifications/read/0 (id=0)	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	SERVER ERROR 500
High	NR-B02 03-NotificationsAPI	PATCH /notifications/read/-1 (id=-1)	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	SERVER ERROR 500
High	NR-B03 03-NotificationsAPI	PATCH /notifications/read/abc (id=abc)	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	SERVER ERROR 500

High	NR-B04 03-NotificationsAPI	PATCH /notifications/read/999999999 (id=999999999 (missing))	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	SERVER ERROR 500
High	NR-B05 03-NotificationsAPI	PATCH /notifications/read/1.5 (id=1.5)	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	SERVER ERROR 500
High	NR-B06 03-NotificationsAPI	PATCH /notifications/read/99999999999999999 (id beyond int range)	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	SERVER ERROR 500
High	CHH-B-abc 04-ChatAPI	GET /chats/abc (non-numeric room)	HTTP 400/404, never 500	HTTP 500	SERVER ERROR 500
High	IT-B-abc 04-ChatAPI	GET /inbox/abc (id=abc)	HTTP 400/403/404, never 500	HTTP 500	SERVER ERROR 500
High	SM-E02 04-ChatAPI	Script tag in a message is stored/escaped (XSS hygiene)	<script> removed or escaped	HTTP 201 stored="[CL-04 1014101612] <script>alert(1)</script> bold"	RAW <script> STORED — FE must sanitise on render; security note
High	RD-B01 04-ChatAPI	PATCH /inbox/abc/read	HTTP 400/404, never 500	HTTP 500	SERVER ERROR 500
High	XR-04 05-RolesPermissions	Customer → GET /teams is refused (COMPANIES read)	HTTP 401/403 (404 acceptable)	HTTP 200 msg="SUCCESS"	ACCESS ALLOWED — permission gap
High	OB-E01 06-Onboarding	HTML with onerror in About me is sanitised	onerror stripped	HTTP 200 stored="<img src=x onerror=alert(1)>"	RAW HTML STORED — FE must sanitise on render (security note)
High	OB-E03 06-Onboarding	APPROVED engineer can call submitProfile again (status flips to SUBMITTED)	HTTP 400/409 (not re-submitted)	HTTP 200 "SUCCESS" — status became SUBMITTED (observed 2026-10-13, account restored by admin)	Server-side guard missing: an approved profile can be re-submitted by calling the API directly; the UI hides the route but the backend allows it.
Medium	OTP-N08 01-AuthAPI	Resend OTP for an already-verified email	HTTP 400/404/409 (should not re-verify)	HTTP 200 msg="Verification otp sent"	informational if 200: server re-sends OTP to a verified user
Medium	SU-N11 01-AuthAPI	Signup rejected: password with space	HTTP 400	HTTP 201 msg="Verification email sent"	
Medium	SU-N12 01-AuthAPI	Signup rejected: invalid referral source	HTTP 400	HTTP 201 msg="Verification email sent"	
Medium	SU-B06 01-AuthAPI	Signup rejected: first name 300 chars	HTTP 400	HTTP 201 msg="Verification email sent"	
Medium	FP-N02 01-AuthAPI	Forgot password: invalid email format	HTTP 400	HTTP 201 body={"sent":true,"message":"If an account exists for this email, a reset link has been sent."}	
Medium	RP-N01 01-AuthAPI	Reset password rejected: invalid token	HTTP 400/401/404	HTTP 300 msg="Invalid link"	
Medium	RP-B01 01-AuthAPI	Reset password rejected: token 63 chars	HTTP 400/401/404	HTTP 300 msg="Invalid link"	
Medium	RP-B02 01-AuthAPI	Reset password rejected: token 65 chars	HTTP 400/401/404	HTTP 300 msg="Invalid link"	
Medium	NB-B-messages--1 03-NotificationsAPI	GET /notifications/messages limit=-1	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=98 (412 notifications)	server accepted limit=-1 and returned 98 items (should probably be 400 or default 10)
Medium	NB-B-messages-1.5 03-NotificationsAPI	GET /notifications/messages limit=1.5	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=1 (4 notifications)	
Medium	NRM-N01 03-NotificationsAPI	PATCH /notifications/messages/readAll/undefined (chatroomId=undefined (FE bug value → NaN))	HTTP 400/404, never 500	HTTP 200 msg="All Messages notifications marked as read for 27 NaN"	
Medium	NRM-B02 03-NotificationsAPI	PATCH /notifications/messages/readAll/abc (chatroomId=abc)	HTTP 400/404, never 500	HTTP 200 msg="All Messages notifications marked as read for 27 NaN"	
Medium	NT-N01 03-NotificationsAPI	PATCH /users/deviceToken with empty body is rejected	HTTP 400	HTTP 200 msg="SUCCESS"	accepts empty body (clears token?) — informational
Medium	AT-N01 04-ChatAPI	POST /chats/files without files is rejected (not 500)	HTTP 400	HTTP 201 msg="{"fileIds":[]}"	accepted an upload with no files (GAP: no limits)
Medium	AT-E01 04-ChatAPI	Uploading an .exe as a chat file	HTTP 400 (type rejected) — GAP if accepted	HTTP 201 body={"data":[3829],"message":"SUCCESS"}	CONFIRMED GAP: executable accepted as chat attachment
Medium	XR-06 05-RolesPermissions	Customer → POST /offers/route is refused (WORK_ORDERS write)	HTTP 401/403 (404 acceptable)	HTTP 400 msg="Select at least one engineer to route."	validation ran before the permission check (informational)
Medium	XR-12 05-RolesPermissions	Customer → GET /users/engineer/submitProfile is refused (FREELANCER only)	HTTP 401/403 (404 acceptable)	HTTP 400 msg="Complete these required fields before submitting your profile: Nationality, ID document, ABN or Foreign Contractor declaration, Hourly, Half Day & Full Day rates, Resume"	validation ran before the permission check (informational)
Medium	OB-B02 06-Onboarding	About me 1001 chars rejected by the server	HTTP 400	HTTP 200 msg="SUCCESS"	SERVER ACCEPTS >1000 chars — only the UI enforces the limit

Medium	`OB-N02` 06-Onboarding	PATCH users/address with empty body is rejected	HTTP 400	HTTP 200 msg="SUCCESS"	
Medium	`OB-N03` 06-Onboarding	PATCH users/address with blank country/city/state is rejected	HTTP 400	HTTP 200 msg="SUCCESS"	SERVER ACCEPTED blank address fields (UI-only validation) — reverted
Low	`URP-N07` 02-AuthUI	Submitting with an invalid token shows "Invalid link" and "Back to Forgot Password" link	error status + "Invalid link" + back link	HTTP 300 text=at least one number Enter at least one symbol (Such as !@#$%^&) Create New Password Back to Forgot Password Invalid link	API answers HTTP 300 for an invalid token — should be 400/404 (finding)
Low	`NB-B-messages-0` 03-NotificationsAPI	GET /notifications/messages limit=0	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=0 (0 notifications)	server accepted limit=0 and returned 0 items (should probably be 400 or default 10)
Low	`NB-B-messages-abc` 03-NotificationsAPI	GET /notifications/messages limit=abc	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=0 (0 notifications)	server accepted limit=abc and returned 0 items (should probably be 400 or default 10)
Low	`NB-B-others-abc` 03-NotificationsAPI	GET /notifications/others limit=abc	HTTP 200/400; items ≤ limit	HTTP 200 items=0 (0 notifications)	server accepted limit=abc and returned 0 items (should probably be 400 or default 10)
Low	`IN-B-0` 04-ChatAPI	GET /inbox limit=0	HTTP 200/400, rows ≤ min(limit,50)	HTTP 200 rows=15	accepted limit=0, returned 15
Low	`IN-B--1` 04-ChatAPI	GET /inbox limit=-1	HTTP 200/400, rows ≤ min(limit,50)	HTTP 200 rows=1	accepted limit=-1, returned 1
Low	`IN-B-abc` 04-ChatAPI	GET /inbox limit=abc	HTTP 200/400, rows ≤ min(limit,50)	HTTP 200 rows=15	accepted limit=abc, returned 15
Low	`OB-E02` 06-Onboarding	ABN "abc" (no format validation per checklist)	informational	HTTP 200 msg="SUCCESS"	server accepts any ABN text — consider 11-digit validation

4. How to read the IDs

- **-P** = positive (normal) case, **-N** = negative (wrong input), **-E** = edge case, **-B** = boundary value.
- LG login · SU sign-up · OTP one-time code · FP forgot password · RP reset password · SE/RT/LO session, refresh, logout · UL/UR/UF/URP/UO/UG/UT/UC UI screens · NA/NB/NR/NRA notifications · CU/IN/CHH/IT/SM/RD/RA chat & inbox · PM/XR/UI roles · OB onboarding.
- Every browser check saved a screenshot under evidence/Checklist/<area>/run-<stamp>/; API checks saved results.json in the same folder.

Authentication API (91 pass / 12 fail)

ID	Kind	What we checked	What should happen	What happened	Result
LG-P-Admin	Normal use (should work)	Login works for Admin	200/201 + accessToken + refreshToken	HTTP 201 token=true refresh=true role=ADMIN	PASS
LG-P-Admin-role	Normal use (should work)	Login response role for Admin	ADMIN	ADMIN	PASS
LG-P-Admin-perm	Normal use (should work)	Permissions attached at login for Admin	array of {module,isRead,...}	permissions=6	PASS
LG-P-Admin-nopw	Wrong input (should be refused)	Login response does not leak password hash (Admin)	no "password" field in body	absent	PASS
LG-P-TechDomain User	Normal use (should work)	Login works for TechDomain User	200/201 + accessToken + refreshToken	HTTP 201 token=true refresh=true role=CLIENT	PASS
LG-P-TechDomain User-role	Normal use (should work)	Login response role for TechDomain User	CLIENT	CLIENT	PASS
LG-P-TechDomain User-perm	Normal use (should work)	Permissions attached at login for TechDomain User	array of {module,isRead,...}	permissions=28	PASS
LG-P-TechDomain User-nopw	Wrong input (should be refused)	Login response does not leak password hash (TechDomain User)	no "password" field in body	absent	PASS
LG-P-Individual Engineer	Normal use (should work)	Login works for Individual Engineer	200/201 + accessToken + refreshToken	HTTP 201 token=true refresh=true role=FREELANCER	PASS
LG-P-Individual Engineer-role	Normal use (should work)	Login response role for Individual Engineer	FREELANCER	FREELANCER	PASS
LG-P-Individual Engineer-perm	Normal use (should work)	Permissions attached at login for Individual Engineer	array of {module,isRead,...}	permissions=2	PASS
LG-P-Individual Engineer-nopw	Wrong input (should be refused)	Login response does not leak password hash (Individual Engineer)	no "password" field in body	absent	PASS
LG-P-Vendor	Normal use (should work)	Login works for Vendor	200/201 + accessToken + refreshToken	HTTP 201 token=true refresh=true role=FREELANCER	PASS
LG-P-Vendor-role	Normal use (should work)	Login response role for Vendor	FREELANCER	FREELANCER	PASS
LG-P-Vendor-perm	Normal use (should work)	Permissions attached at login for Vendor	array of {module,isRead,...}	permissions=3	PASS
LG-P-Vendor-nopw	Wrong input (should be refused)	Login response does not leak password hash (Vendor)	no "password" field in body	absent	PASS
LG-P-Vendor Engineer	Normal use (should work)	Login works for Vendor Engineer	200/201 + accessToken + refreshToken	HTTP 201 token=true refresh=true role=FREELANCER	PASS
LG-P-Vendor Engineer-role	Normal use (should work)	Login response role for Vendor Engineer	FREELANCER	FREELANCER	PASS
LG-P-Vendor Engineer-perm	Normal use (should work)	Permissions attached at login for Vendor Engineer	array of {module,isRead,...}	permissions=2	PASS
LG-P-Vendor Engineer-nopw	Wrong input (should be refused)	Login response does not leak password hash (Vendor Engineer)	no "password" field in body	absent	PASS
LG-P-Customer	Normal use (should work)	Login works for Customer	200/201 + accessToken + refreshToken	HTTP 201 token=true refresh=true role=CUSTOMER	PASS
LG-P-Customer-role	Normal use (should work)	Login response role for Customer	CUSTOMER	CUSTOMER	PASS
LG-P-Customer-perm	Normal use (should work)	Permissions attached at login for Customer	array of {module,isRead,...}	permissions=10	PASS
LG-P-Customer-nopw	Wrong input (should be refused)	Login response does not leak password hash (Customer)	no "password" field in body	absent	PASS
LG-P-Customer User	Normal use (should work)	Login works for Customer User	200/201 + accessToken + refreshToken	HTTP 201 token=true refresh=true role=CUSTOMER	PASS

LG-P-Customer User-role	Normal use (should work)	Login response role for Customer User	CUSTOMER	CUSTOMER	**PASS**
LG-P-Customer User-perm	Normal use (should work)	Permissions attached at login for Customer User	array of {module,isRead,...}	permissions=9	**PASS**
LG-P-Customer User-nopw	Wrong input (should be refused)	Login response does not leak password hash (Customer User)	no "password" field in body	absent	**PASS**
LG-N01	Wrong input (should be refused)	Login rejected: wrong password	HTTP 401/404, no token, generic message	HTTP 401 token=false msg="Invalid email or password"	**PASS**
LG-N02	Wrong input (should be refused)	Login rejected: unregistered email	HTTP 401/404, no token, generic message	HTTP 404 token=false msg="Invalid email or password"	**PASS**
LG-N03	Wrong input (should be refused)	Login rejected: email upper-cased + correct password Note: informational: case/space handling	HTTP 200/201/401/404, no token, generic message	HTTP 201 token=true msg="Login successfully"	**PASS**
LG-N04	Wrong input (should be refused)	Login rejected: empty email	HTTP 400/401/404, no token, generic message	HTTP 401 token=false msg="Unauthorized"	**PASS**
LG-N05	Wrong input (should be refused)	Login rejected: empty password	HTTP 400/401/404, no token, generic message	HTTP 401 token=false msg="Unauthorized"	**PASS**
LG-N06	Wrong input (should be refused)	Login rejected: missing body	HTTP 400/401/404, no token, generic message	HTTP 401 token=false msg="Unauthorized"	**PASS**
LG-N07	Wrong input (should be refused)	Login rejected: invalid email format	HTTP 400/401/404, no token, generic message	HTTP 404 token=false msg="Invalid email or password"	**PASS**
LG-N08	Wrong input (should be refused)	Login rejected: email with leading/trailing spaces Note: informational: case/space handling	HTTP 200/201/400/401/404, no token, generic message	HTTP 201 token=true msg="Login successfully"	**PASS**
LG-N09	Wrong input (should be refused)	Login rejected: SQL injection in email	HTTP 400/401/404, no token, generic message	HTTP 404 token=false msg="Invalid email or password"	**PASS**
LG-N10	Limit value (min / max / just outside)	Login rejected: password 1 char	HTTP 400/401/404, no token, generic message	HTTP 401 token=false msg="Invalid email or password"	**PASS**
LG-N11	Limit value (min / max / just outside)	Login rejected: password 500 chars	HTTP 400/401/404/413, no token, generic message	HTTP 401 token=false msg="Invalid email or password"	**PASS**
LG-N12	Limit value (min / max / just outside)	Login rejected: email 320 chars	HTTP 400/401/404, no token, generic message	HTTP 404 token=false msg="Invalid email or password"	**PASS**
LG-N13	Limit value (min / max / just outside)	Login rejected: email as number	HTTP 400/401/404, no token, generic message	HTTP 500 token=false msg="Internal server error"	**FAIL**
LG-N14	Limit value (min / max / just outside)	Login rejected: null fields	HTTP 400/401/404, no token, generic message	HTTP 401 token=false msg="Unauthorized"	**PASS**
LG-N15	Unusual situation	8 quick failed logins: server stays consistent (no 500) Note: no rate limiting observed (informational)	all 401/404/429	401,401,401,401,401,401,401,401	**PASS**
SU-P01	Normal use (should work)	Technician signup with valid data is accepted	200/201	HTTP 201 msg="Verification email sent"	**PASS**
LG-E01	Unusual situation	Login of unverified email returns 307 (OTP re-sent)	HTTP 307, no token	HTTP 307 token=false	**PASS**
SU-N01	Wrong input (should be refused)	Signup again with same unverified email → 403 (go to OTP)	HTTP 403	HTTP 500 msg="User already exists with this email, OTP code sent"	**FAIL**
OTP-N01	Limit value (min / max / just outside)	OTP verify rejected: 5 digits	HTTP 400/401/404, no token	HTTP 404 token=false msg="Invalid OTP"	**PASS**
OTP-N02	Limit value (min / max / just outside)	OTP verify rejected: 7 digits	HTTP 400/401/404, no token	HTTP 404 token=false msg="Invalid OTP"	**PASS**

OTP-N03	Wrong input (should be refused)	OTP verify rejected: letters	HTTP 400/401/404, no token	HTTP 404 token=false msg="Invalid OTP"	**PASS**
OTP-N04	Wrong input (should be refused)	OTP verify rejected: wrong 6 digits	HTTP 400/401/404, no token	HTTP 404 token=false msg="Invalid OTP"	**PASS**
OTP-N05	Wrong input (should be refused)	OTP verify rejected: empty code	HTTP 400/401/404, no token	HTTP 400 token=false msg="code should not be empty"	**PASS**
OTP-N06	Wrong input (should be refused)	OTP verify rejected: code as number	HTTP 400/401/404, no token	HTTP 500 token=false msg="Something went wrong"	**FAIL**
OTP-P01	Normal use (should work)	Resend OTP for an unverified email	HTTP 200/201	HTTP 200 msg="Verification otp sent"	**PASS**
OTP-N07	Wrong input (should be refused)	Resend OTP for unknown email is rejected	HTTP 400/404	HTTP 404 msg="User not found"	**PASS**
OTP-N08	Unusual situation	Resend OTP for an already-verified email Note: informational if 200: server re-sends OTP to a verified user	HTTP 400/404/409 (should not re-verify)	HTTP 200 msg="Verification otp sent"	**FAIL**
SU-N02	Wrong input (should be refused)	Signup rejected: existing verified email	HTTP 400/403/409	HTTP 500 msg="User already exists."	**FAIL**
SU-N03	Wrong input (should be refused)	Signup rejected: invalid email	HTTP 400	HTTP 400 msg="email must be an email"	**PASS**
SU-N04	Wrong input (should be refused)	Signup rejected: missing email	HTTP 400	HTTP 400 msg="email must be an email | email should not be empty"	**PASS**
SU-N05	Wrong input (should be refused)	Signup rejected: missing password	HTTP 400	HTTP 400 msg="Password must be 8-20 characters and include upper and lower case letters, a number and a symbol | password must be longer than or equal to 8 characters | password should not be empty | password must be a string"	**PASS**
SU-N06	Wrong input (should be refused)	Signup rejected: missing first name	HTTP 400	HTTP 400 msg="First name must only contain letters | firstName should not be empty | firstName must be a string"	**PASS**
SU-N07	Wrong input (should be refused)	Signup rejected: first name with digits	HTTP 400	HTTP 400 msg="First name must only contain letters"	**PASS**
SU-B01	Limit value (min / max / just outside)	Signup rejected: password 7 chars (below min 8)	HTTP 400	HTTP 400 msg="Password must be 8-20 characters and include upper and lower case letters, a number and a symbol | password must be longer than or equal to 8 characters"	**PASS**
SU-B02	Limit value (min / max / just outside)	Signup accepted: password 8 chars (min)	HTTP 200/201	HTTP 201 msg="Verification email sent"	**PASS**
SU-B03	Limit value (min / max / just outside)	Signup accepted: password 20 chars (max)	HTTP 200/201	HTTP 201 msg="Verification email sent"	**PASS**
SU-B04	Limit value (min / max / just outside)	Signup rejected: password 21 chars (above max 20)	HTTP 400	HTTP 400 msg="Password must be 8-20 characters and include upper and lower case letters, a number and a symbol"	**PASS**
SU-N08	Wrong input (should be refused)	Signup rejected: password without uppercase	HTTP 400	HTTP 400 msg="Password must be 8-20 characters and include upper and lower case letters, a number and a symbol"	**PASS**
SU-N09	Wrong input (should be refused)	Signup rejected: password without number	HTTP 400	HTTP 400 msg="Password must be 8-20 characters and include upper and lower case letters, a number and a symbol"	**PASS**
SU-N10	Wrong input (should be refused)	Signup rejected: password without symbol	HTTP 400	HTTP 400 msg="Password must be 8-20 characters and include upper and lower case letters, a number and a symbol"	**PASS**
SU-N11	Wrong input (should be refused)	Signup rejected: password with space	HTTP 400	HTTP 201 msg="Verification email sent"	**FAIL**
SU-N12	Wrong input (should be refused)	Signup rejected: invalid referral source	HTTP 400	HTTP 201 msg="Verification email sent"	**FAIL**

SU-N13	Wrong input (should be refused)	Signup rejected: empty body	HTTP 400	HTTP 400 msg="First name must only contain letters | firstName should not be empty | firstName must be a string | Last name must only contain letters | lastName should not be empty | lastName must be a string | email must be an email | email should not be empty | phoneNumber should not be empty | phoneNumber must be a string | Password must be 8-20 characters and include upper and lower case lette	**PASS**
SU-B05	Limit value (min / max / just outside)	Signup accepted: first name 1 char Note: informational boundary	HTTP 200/201/400	HTTP 201 msg="Verification email sent"	**PASS**
SU-B06	Limit value (min / max / just outside)	Signup rejected: first name 300 chars	HTTP 400	HTTP 201 msg="Verification email sent"	**FAIL**
FP-P01	Normal use (should work)	Forgot password: registered email	HTTP 200/201	HTTP 201 body={"sent":true,"message":"If an account exists for this email, a reset link has been sent."}	**PASS**
FP-N01	Wrong input (should be refused)	Forgot password: unregistered email (same neutral answer)	HTTP 200/201	HTTP 201 body={"sent":true,"message":"If an account exists for this email, a reset link has been sent."}	**PASS**
FP-N02	Wrong input (should be refused)	Forgot password: invalid email format	HTTP 400	HTTP 201 body={"sent":true,"message":"If an account exists for this email, a reset link has been sent."}	**FAIL**
FP-N03	Wrong input (should be refused)	Forgot password: empty email	HTTP 400	HTTP 400 body={"message":["email should not be empty"],"error":"Bad Request","statusCode":400}	**PASS**
FP-N04	Wrong input (should be refused)	Forgot password: missing body	HTTP 400	HTTP 400 body={"message":["email should not be empty","email must be a string"],"error":"Bad Request","statusCode":400}	**PASS**
FP-B01	Limit value (min / max / just outside)	Forgot password: email 320 chars	HTTP 200/201/400	HTTP 201 body={"sent":true,"message":"If an account exists for this email, a reset link has been sent."}	**PASS**
FP-E01	Unusual situation	Registered vs unregistered email give the same response (no enumeration)	same status + same message	201/"If an account exists for this email, a reset link has been sent." vs 201/"If an account exists for this email, a reset link has been sent."	**PASS**
FP-E02	Unusual situation	Second request inside cooldown is handled (sent=false or 429) Note: informational if server simply re-sends. Automated call hit a transient ECONNRESET on 2026-10-14; re-checked manually with curl (2 calls back to back, both 201 sent=true), same as 13 Oct.	HTTP 200 with sent!=true/cooldownRemaining, or 429	HTTP 201 body={"sent":true,"message":"If an account exists for this email, a reset link has been sent."}	**PASS**
RP-N01	Wrong input (should be refused)	Reset password rejected: invalid token	HTTP 400/401/404	HTTP 300 msg="Invalid link"	**FAIL**
RP-N02	Wrong input (should be refused)	Reset password rejected: empty token	HTTP 400/401/404	HTTP 400 msg="resetToken should not be empty"	**PASS**
RP-N03	Wrong input (should be refused)	Reset password rejected: missing token	HTTP 400/401/404	HTTP 400 msg="resetToken should not be empty | resetToken must be a string"	**PASS**
RP-B01	Limit value (min / max / just outside)	Reset password rejected: token 63 chars	HTTP 400/401/404	HTTP 300 msg="Invalid link"	**FAIL**
RP-B02	Limit value (min / max / just outside)	Reset password rejected: token 65 chars	HTTP 400/401/404	HTTP 300 msg="Invalid link"	**FAIL**
RP-B03	Limit value (min / max / just outside)	Reset password rejected: password 7 chars + bad token	HTTP 400/401/404	HTTP 400 msg="Password must be 8-20 characters and include upper and lower case letters, a number and a symbol | newPassword must be longer than or equal to 8 characters"	**PASS**
RP-N04	Wrong input (should be refused)	Reset password rejected: weak password (no symbol) + bad token	HTTP 400/401/404	HTTP 400 msg="Password must be 8-20 characters and include upper and lower case letters, a number and a symbol"	**PASS**
RP-N05	Wrong input (should be refused)	Reset password rejected: empty body	HTTP 400/401/404	HTTP 400 msg="resetToken should not be empty | resetToken must be a string | Password must be 8-20 characters and include upper and lower case letters, a number and a symbol | newPassword must be longer than or equal to 8 characters | newPassword should not be empty | newPassword must be a string"	**PASS**
SE-N01	Wrong input (should be refused)	GET auth/session without token → 401	HTTP 401	HTTP 401	**PASS**
CNP-N01	Wrong input (should be refused)	PUT auth/createNewPassword without token → 401	HTTP 401	HTTP 401	**PASS**
SE-N02	Wrong input (should be	Protected endpoint (users/me) without token → 401	HTTP 401	HTTP 401	**PASS**

	refused)				
SE-N03	Wrong input (should be refused)	Malformed bearer token → 401	HTTP 401	HTTP 401	**PASS**
SE-P01	Normal use (should work)	GET auth/session with a valid token returns the user	HTTP 200 + user	HTTP 200 keys=data	**PASS**
SE-P02	Normal use (should work)	GET users/me with a valid token	HTTP 200	HTTP 200	**PASS**
RT-P01	Normal use (should work)	Refresh token exchange returns new tokens	HTTP 200/201 + accessToken + refreshToken	HTTP 201 access=true refresh=true	**PASS**
RT-N01	Wrong input (should be refused)	Refresh with a bogus refresh token is rejected	HTTP 401/403/400	HTTP 401 msg="Invalid or expired refresh token"	**PASS**
RT-N02	Wrong input (should be refused)	Refresh with empty body is rejected	HTTP 400/401	HTTP 400	**PASS**
RT-E01	Unusual situation	Re-using an already-exchanged refresh token Note: OLD REFRESH TOKEN STILL ACCEPTED (no rotation) — security note	HTTP 401/403 (rotated) — informational if accepted	HTTP 201	**PASS**
SE-E01	Unusual situation	A new login invalidates the previous access token of the same user Note: multi-session: old token stays valid after a new login	informational (401 = single session, 200 = multi session)	HTTP 200	**PASS**
LO-P01	Normal use (should work)	Logout with refresh token succeeds	HTTP 200/201	HTTP 201 msg="Logout successfully"	**PASS**
LO-E01	Unusual situation	Access token after logout Note: token invalidated	HTTP 401 preferred	HTTP 401	**PASS**
LO-N01	Wrong input (should be refused)	Logout with a bogus refresh token	HTTP 400/401/404 (or 200 idempotent)	HTTP 401	**PASS**

Authentication screens (UI) (89 pass / 0 fail)

ID	Kind	What we checked	What should happen	What happened	Result
UL-01	Normal use (should work)	Login page shows heading "Welcome to Trusted Platform"	text present	present	PASS
UL-02	Normal use (should work)	Email placeholder is "William123@gmail.com"	William123@gmail.com	William123@gmail.com	PASS
UL-03	Normal use (should work)	Password field is masked (type=password)	type=password	password	PASS
UL-04	Normal use (should work)	Eye icon toggles password visibility	type becomes text	text	PASS
UL-05	Wrong input (should be refused)	No Google / Apple social buttons on login	absent	absent	PASS
UL-06	Normal use (should work)	App Store badge links to the Tech Domain iOS app	href contains id6742741677	https://apps.apple.com/us/app/tech-domain/id6742741677	PASS
UL-07	Normal use (should work)	Google Play badge links to com.techdomain.tdm	href contains com.techdomain.tdm	https://play.google.com/store/apps/details?id=com.techdomain.tdm&pli=1	PASS
UL-08	Normal use (should work)	"Forgot Password?" link goes to /forgotPassword	/forgotPassword	/forgotPassword	PASS
UL-09	Normal use (should work)	"SignUp" link goes to /getStarted	/getStarted	/getStarted	PASS
UL-10	Normal use (should work)	"Remember Me" checkbox is shown	present	present	PASS
UL-11	Wrong input (should be refused)	Empty email on submit → "Email is required"	Email is required	shown	PASS
UL-12	Wrong input (should be refused)	Empty password on submit → "Password is required"	Password is required	shown	PASS
UL-13	Wrong input (should be refused)	Invalid email format → "Enter a valid email"	Enter a valid email	shown	PASS
UL-14	Unusual situation	Validation does not run on change/blur (only on submit)	no error before submit	no error	PASS
UL-15	Wrong input (should be refused)	Wrong password → red banner "Invalid Email/Password" and stays on /login	banner shown, url /login	HTTP 401 banner=true url=https://dev.techdomain.com.au/login	PASS
UL-16	Limit value (min / max / just outside)	320-char email: page handles it without crashing	error shown, page alive	error shown	PASS
UL-17	Normal use (should work)	Engineer login lands on /jobs	/jobs	https://dev.techdomain.com.au/jobs	PASS
UL-18	Unusual situation	Logged-in user opening /login is redirected away	not /login	https://dev.techdomain.com.au/jobs	PASS
UL-19	Unusual situation	Logged-in user opening /getStarted is redirected away	not /getStarted	https://dev.techdomain.com.au/jobs	PASS
UL-20	Wrong input (should be refused)	Unknown URL while logged in shows "Unauthorized"	Unauthorized heading	shown	PASS
UL-21	Wrong input (should be refused)	APPROVED engineer cannot open /completeYourProfile (route removed → Unauthorized)	Unauthorized / redirected	https://dev.techdomain.com.au/completeYourProfile text=Analytics Job 0 11 Unauthorized	PASS
UL-22	Normal use (should work)	Logout from the top bar returns to /login	/login	https://dev.techdomain.com.au/login	PASS
UL-23	Wrong input (should be refused)	After logout a protected URL redirects to /getStarted	/getStarted	https://dev.techdomain.com.au/getStarted	PASS
UR-01	Wrong input (should be refused)	Logged-out visit to /workBoard/workOrders redirects to /getStarted	/getStarted	https://dev.techdomain.com.au/getStarted	PASS
UR-02	Wrong input (should be refused)	Logged-out visit to /admin/review/engineers redirects to /getStarted	/getStarted	https://dev.techdomain.com.au/getStarted	PASS
UR-03	Wrong input (should be refused)	Logged-out visit to /jobs redirects to /getStarted	/getStarted	https://dev.techdomain.com.au/getStarted	PASS
UR-04	Wrong input (should be refused)	Logged-out visit to /some/unknown-1014101344 redirects to /getStarted	/getStarted	https://dev.techdomain.com.au/getStarted	PASS
UR-05	Normal use (should work)	/not-found shows "No User Linked :("	No User Linked	No User Linked :(	PASS
UR-06	Normal use (should work)	/unauthorized renders "Unauthorized"	Unauthorized	Unauthorized	PASS
UR-07	Unusual situation	/ while logged out: placeholder or redirect to Get Started Note: checklist says placeholder, site shows Get Started — doc out of date (fine)	FlightboardPage text OR /getStarted	https://dev.techdomain.com.au/getStarted text=Join Tech Domain Connecting Businesses with Skilled Field Te	PASS
UF-01	Normal use (should work)	Forgot page heading and helper text	both texts	Forgot Password ? Provide your Registered E-mail address and we'll send you a password reset link. Email Email Send recovery link Return to Login	PASS
UF-02	Normal use (should work)	Button label "Send recovery link"	Send recovery link	Send recovery link	PASS
UF-03	Wrong input (should be refused)	Invalid email validates before submit → "Enter a valid email"	Enter a valid email	shown	PASS
UF-04	Wrong input (should be refused)	Cleared email → "Email is required"	Email is required	shown	PASS
UF-05	Wrong input (should be refused)	Unregistered email → neutral success view (no enumeration)	"A password recovery email … has been sent" and no "not registered"	HTTP 201 text=If an account exists for the email below, a password recovery email with instructions has been sent. Please follow the instructions to reset the password. nobod	PASS

UF-06	Normal use (should work)	Success view shows the entered email and "Return to login"	email + Return to login	If an account exists for the email below, a password recovery email with instructions has been sent. Please follow the instructions to reset the password. nobody-1014101344@mailinator.com Return to lo	PASS
UF-07	Unusual situation	Back to form shows disabled "Resend in m:ss" cooldown	Resend in 0:5x	Resend in 0:58	PASS
UF-08	Unusual situation	Reload during cooldown keeps the countdown	Resend in m:ss	Resend in 0:55	PASS
URP-01	Normal use (should work)	Reset page heading "Reset Password" + "Please Enter your new password"	both texts	Reset Password Please Enter your new password New Password * New Password * Confirm Password * Confirm Password * Enter from 8 to 20 cha	PASS
URP-02	Normal use (should work)	Two password fields (New + Confirm)	2	2	PASS
URP-B01	Limit value (min / max / just outside)	New password validation: 7 chars → "at least 8 characters"	/at least 8 characters/i	message shown	PASS
URP-N01	Wrong input (should be refused)	New password validation: no uppercase	/uppercase/i	message shown	PASS
URP-N02	Wrong input (should be refused)	New password validation: no lowercase	/lowercase/i	message shown	PASS
URP-N03	Wrong input (should be refused)	New password validation: no number	/number/i	message shown	PASS
URP-N04	Wrong input (should be refused)	New password validation: no symbol	/symbol/i	message shown	PASS
URP-N05	Wrong input (should be refused)	New password validation: contains space	/spaces/i	message shown	PASS
URP-B02	Limit value (min / max / just outside)	8-char valid password → no rule error	no error	no error	PASS
URP-B03	Limit value (min / max / just outside)	30-char password: no max-length rule on reset (checker still says 8–20) Note: inconsistency between form rule and checker text — product to decide	no "at most 20" error (documented gap)	no max error	PASS
URP-N06	Wrong input (should be refused)	Confirm mismatch → "Passwords must match"	Passwords must match	shown	PASS
URP-N07	Wrong input (should be refused)	Submitting with an invalid token shows "Invalid link" and "Back to Forgot Password" link Note: API answers HTTP 300 for an invalid token — should be 400/404 (finding)	error status + "Invalid link" + back link	HTTP 300 text=at least one number Enter at least one symbol (Such as !@#$%^&) Create New Password Back to Forgot Password Invalid link	PASS
UO-01	Normal use (should work)	OTP page shows "Email Verification" and the email	Email Verification + email	Email Verification We have sent a (OTP) to your email "cl02-otp-1014101344@mailinator.com" Verify 0:59	PASS
UO-02	Normal use (should work)	Six single-digit OTP boxes	6	6	PASS
UO-03	Wrong input (should be refused)	"Verify" disabled until 6 digits entered	disabled	disabled	PASS
UO-04	Normal use (should work)	Countdown timer starts near 1:00	0:5x or 1:00	0:59	PASS
UO-B01	Limit value (min / max / just outside)	5 digits entered → Verify still disabled	disabled	disabled	PASS
UO-B02	Limit value (min / max / just outside)	6 digits entered → Verify enabled	enabled	enabled	PASS
UO-N01	Wrong input (should be refused)	Letters are not accepted in OTP boxes	box stays numeric/empty	value="2"	PASS
UO-N02	Wrong input (should be refused)	Wrong code → server error shown as toast, stays on /otp	HTTP 4xx, still /otp	HTTP 404 url=https://dev.techdomain.com.au/otp?email=cl02-otp-1014101344%40mailinator.com text=Email Verification We have sent a (OTP) to your email "cl02-otp-1014101344@mailinator.com" Verify 0:55 Invalid OTP	PASS
UG-01	Normal use (should work)	Get Started shows "Join Tech Domain" + both cards	Looking for Work + Looking for Technicians	Join Tech Domain Connecting Businesses with Skilled Field Technicians—When and Where They're Needed Connect with opportunities or skilled field technicians—all in one platform Looking for Work For tec	PASS
UG-02	Wrong input (should be refused)	"Get Started" button disabled until a card is selected	disabled	disabled	PASS
UG-03	Wrong input (should be refused)	No social sign-up buttons	absent	absent	PASS
UG-04	Normal use (should work)	Selecting "Looking for Work" adds ?signup=TECH to the URL	signup=TECH	https://dev.techdomain.com.au/getStarted?signup=TECH	PASS
UG-05	Normal use (should work)	TECH → /techSignup	/techSignup	https://dev.techdomain.com.au/techSignup	PASS
UG-06	Unusual situation	?signup=CLIENT pre-selects the card (button enabled)	enabled	enabled	PASS
UG-07	Normal use (should work)	CLIENT → /customerSignup (not /clientSignup)	/customerSignup	https://dev.techdomain.com.au/customerSignup	PASS
UG-08	Wrong input (should be refused)	Invalid ?ticket= shows the expired-link message and disables the button	sign-in link has expired	Join Tech Domain One last step before we set you up Your Google sign-in link has expired. Please sign in again. Looking for Work For technicians & Service Providers/Vendors Find work. Build connection	PASS
UT-01	Normal use (should work)	Heading "Let's Create Your Account" and provider-type question	both	Let's Create Your Account Connect. Work. Grow. Create your professional profile and access field service opportunities that match your expertise and availability. How do you provide services? * Indepe	PASS
UT-02	Unusual situation	Form hidden until a provider type is chosen	no password field yet	password fields=0	PASS

UT-03	Normal use (should work)	Choosing "Independent Technician" reveals the form	password field visible	password fields=1	**PASS**
UT-04	Normal use (should work)	"Company or Team" shows "Your company name *"	company name field	shown	**PASS**
UT-05	Wrong input (should be refused)	First name filters digits while typing	John	John	**PASS**
UT-B01	Limit value (min / max / just outside)	First name accepts 100 letters (no max on UI — informational) Note: no client max length	informational	length=100	**PASS**
UT-06	Wrong input (should be refused)	"Create Account" disabled until required fields + consent	disabled	disabled	**PASS**
UT-07	Normal use (should work)	PasswordChecker appears once password has a value	checker text (8 to 20 characters)	shown	**PASS**
UT-08	Normal use (should work)	"Privacy Statement" opens the Privacy Policy modal	opened	opened	**PASS**
UT-09	Wrong input (should be refused)	All fields filled but consent unchecked → still disabled	disabled	disabled	**PASS**
UT-10	Normal use (should work)	Consent checked + all fields → "Create Account" enabled	enabled	enabled	**PASS**
UT-11	Wrong input (should be refused)	Signup with an existing verified email → error toast, stays on signup	HTTP 4xx + toast	HTTP 500 text=Let's Create Your Account Connect. Work. Grow. Create your professional profile and access field service opportunities t url=https://dev.techdomain.com.au/techSignup	**PASS**
UC-01	Normal use (should work)	Customer sign-up form fields present	Company Name + Annual Revenue + Where did you hear	Let's Create Your Account Find the Right Technicians. Get Work Done with Confidence. Create your Tech Domain account to connect with skilled field technicians and manage your service requirements from	**PASS**
UC-02	Wrong input (should be refused)	"Create Account" disabled initially	disabled	disabled	**PASS**
UC-03	Wrong input (should be refused)	First name letters only (digits filtered)	Mary	Mary	**PASS**
UC-04	Wrong input (should be refused)	Blur with empty first name → "First name is required"	First name is required	shown	**PASS**
UC-B01	Limit value (min / max / just outside)	7-char password → "at least 8 characters"	at least 8	shown	**PASS**
UC-B02	Limit value (min / max / just outside)	21-char password → "at most 20 characters"	at most 20	shown	**PASS**
UC-B03	Limit value (min / max / just outside)	20-char password → accepted (no length error)	no length error	no length error	**PASS**
UCL-01	Unusual situation	/clientSignup is reachable by direct URL	Create your account + Company Name	Let's Create your account Looking to hire skilled technicians and post work orders? Just complete the form below to get started First Name * First Name * Last N	**PASS**
UCL-02	Wrong input (should be refused)	Client form: Create Account disabled only while consent unchecked	disabled before consent	disabled	**PASS**
UCL-03	Unusual situation	Client form: consent checked → button enabled even with empty fields (validation on submit)	enabled	enabled	**PASS**
UCL-04	Wrong input (should be refused)	Client form empty submit shows required-field messages	First name is required etc.	Let's Create your account Looking to hire skilled technicians and post work orders? Just complete the form below to get started First Name * First Name * First name is required Last Name * Last Name *	**PASS**

Notifications API (§1.3) (52 pass / 14 fail)

ID	Kind	What we checked	What should happen	What happened	Result
NA-N01	Wrong input (should be refused)	GET /notifications without token → 401	HTTP 401	HTTP 401	PASS
NA-N02	Wrong input (should be refused)	GET /notifications/messages without token → 401	HTTP 401	HTTP 401	PASS
NA-N03	Wrong input (should be refused)	GET /notifications/others without token → 401	HTTP 401	HTTP 401	PASS
NA-N04	Wrong input (should be refused)	GET /notifications/chat without token → 401	HTTP 401	HTTP 401	PASS
NA-N05	Wrong input (should be refused)	PATCH /notifications/readAll without token → 401	HTTP 401	HTTP 401	PASS
NA-N06	Wrong input (should be refused)	PATCH /notifications/read/1 without token → 401	HTTP 401	HTTP 401	PASS
NA-P01	Normal use (should work)	GET /notifications returns the full list (not paginated)	HTTP 200 + array	HTTP 200 items=557	PASS
NA-P02	Normal use (should work)	MESSAGE notifications come first, then the rest	all MESSAGE before non-MESSAGE	lastMessageIdx=417 firstOtherIdx=418 (n=557)	PASS
NA-P03	Normal use (should work)	Each notification has id/type/title/content/url/isRead/createdAt	fields present	keys=id,receiverId,senderId,workOrderId,chatRoomId,type,title,content,url,isRead,createdAt,updatedAt,sender,referenceId	PASS
NA-E01	Unusual situation	List does not expose receiver.deviceToken	no deviceToken in REST list	absent	PASS
NA-E02	Unusual situation	User never receives a notification about their own action (senderId ≠ me)	0 self-sent	self-sent=0 of 557	PASS
NB-P01-messages	Normal use (should work)	GET /notifications/messages default returns {data, nextCursor, unreadNotificationCount}	shape + ≤10 rooms	HTTP 200 keys=data,nextCursor,unreadNotificationCount items=25 units=10	PASS
NB-B-messages-0	Limit value (min / max / just outside)	GET /notifications/messages limit=0 Note: server accepted limit=0 and returned 0 items (should probably be 400 or default 10)	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=0 (0 notifications)	PASS
NB-B-messages-1	Limit value (min / max / just outside)	GET /notifications/messages limit=1 → 1 item	HTTP 200; rooms ≤ limit	HTTP 200 rooms=1 (4 notifications)	PASS
NB-B-messages-10	Limit value (min / max / just outside)	GET /notifications/messages limit=10 (default)	HTTP 200; rooms ≤ limit	HTTP 200 rooms=10 (25 notifications)	PASS
NB-B-messages-11	Limit value (min / max / just outside)	GET /notifications/messages limit=11	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=11 (28 notifications)	PASS
NB-B-messages-100	Limit value (min / max / just outside)	GET /notifications/messages limit=100	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=99 (418 notifications)	PASS
NB-B-messages-101	Limit value (min / max / just outside)	GET /notifications/messages limit=101	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=99 (418 notifications)	PASS
NB-B-messages--1	Limit value (min / max / just outside)	GET /notifications/messages limit=-1 Note: server accepted limit=-1 and returned 98 items (should probably be 400 or default 10)	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=98 (412 notifications)	FAIL
NB-B-messages-abc	Limit value (min / max / just outside)	GET /notifications/messages limit=abc Note: server accepted limit=abc and returned 0 items (should probably be 400 or default 10)	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=0 (0 notifications)	PASS
NB-B-messages-1.5	Limit value (min / max / just outside)	GET /notifications/messages limit=1.5	HTTP 200/400; rooms ≤ limit	HTTP 200 rooms=1 (4 notifications)	FAIL
NB-C-messages-abc	Unusual situation	GET /notifications/messages cursor=abc	HTTP 200 (empty or page) or 400, never 500	HTTP 200 items=9	PASS
NB-C-messages-0	Unusual situation	GET /notifications/messages cursor=0	HTTP 200 (empty or page) or 400, never 500	HTTP 200 items=9	PASS
NB-C-messages--5	Unusual situation	GET /notifications/messages cursor=-5	HTTP 200 (empty or page) or 400, never 500	HTTP 200 items=9	PASS
NB-C-messages-999999999	Unusual situation	GET /notifications/messages cursor=huge	HTTP 200 (empty or page) or 400, never 500	HTTP 200 items=9	PASS
NB-P02-messages	Normal use (should work)	Second page via nextCursor has no overlap with page 1 (messages)	0 duplicates	HTTP 200 dup=0 n=72	PASS
NB-P01-others	Normal use (should work)	GET /notifications/others default returns {data, nextCursor, unreadNotificationCount}	shape + ≤10 items	HTTP 200 keys=data,nextCursor,unreadNotificationCount items=10 units=10	PASS
NB-B-others-0	Limit value (min / max / just outside)	GET /notifications/others limit=0 Note: SERVER ERROR 500	HTTP 200/400; items ≤ limit	HTTP 500 items=0 (undefined notifications)	FAIL
NB-B-others-1	Limit value (min / max / just outside)	GET /notifications/others limit=1 → 1 item	HTTP 200; items ≤ limit	HTTP 200 items=1 (1 notifications)	PASS
NB-B-others-10	Limit value (min / max / just outside)	GET /notifications/others limit=10 (default)	HTTP 200; items ≤ limit	HTTP 200 items=10 (10 notifications)	PASS
NB-B-others-11	Limit value (min / max / just outside)	GET /notifications/others limit=11	HTTP 200/400; items ≤ limit	HTTP 200 items=11 (11 notifications)	PASS
NB-B-others-100	Limit value (min / max / just outside)	GET /notifications/others limit=100	HTTP 200/400; items ≤ limit	HTTP 200 items=100 (100 notifications)	PASS

NB-B-others-101	Limit value (min / max / just outside)	GET /notifications/others limit=101	HTTP 200/400; items ≤ limit	HTTP 200 items=101 (101 notifications)	PASS
NB-B-others--1	Limit value (min / max / just outside)	GET /notifications/others limit=-1 Note: SERVER ERROR 500	HTTP 200/400; items ≤ limit	HTTP 500 items=0 (undefined notifications)	FAIL
NB-B-others-abc	Limit value (min / max / just outside)	GET /notifications/others limit=abc Note: server accepted limit=abc and returned 0 items (should probably be 400 or default 10)	HTTP 200/400; items ≤ limit	HTTP 200 items=0 (0 notifications)	PASS
NB-B-others-1.5	Limit value (min / max / just outside)	GET /notifications/others limit=1.5	HTTP 200/400; items ≤ limit	HTTP 200 items=1 (1 notifications)	PASS
NB-C-others-abc	Unusual situation	GET /notifications/others cursor=abc	HTTP 200 (empty or page) or 400, never 500	HTTP 200 items=5	PASS
NB-C-others-0	Unusual situation	GET /notifications/others cursor=0	HTTP 200 (empty or page) or 400, never 500	HTTP 200 items=5	PASS
NB-C-others--5	Unusual situation	GET /notifications/others cursor=-5	HTTP 200 (empty or page) or 400, never 500	HTTP 200 items=5	PASS
NB-C-others-999999999	Unusual situation	GET /notifications/others cursor=huge	HTTP 200 (empty or page) or 400, never 500	HTTP 200 items=5	PASS
NB-P02-others	Normal use (should work)	Second page via nextCursor has no overlap with page 1 (others)	0 duplicates	HTTP 200 dup=0 n=10	PASS
NC-P01	Normal use (should work)	GET /notifications/chat returns unread MESSAGE notifications only	HTTP 200; every item type=MESSAGE & isRead=false	HTTP 200 n=2 badTypes=0 read=0	PASS
NR-P01	Normal use (should work)	Mark own notification read	HTTP 200 + isRead true	HTTP 200 isRead=undefined	PASS
NR-E01	Unusual situation	Marking the same notification read twice is idempotent	HTTP 200	HTTP 200	PASS
NR-N01	Wrong input (should be refused)	Marking ANOTHER user's notification read is refused with 404/403 (not 500) Note: CONFIRMED BUG: 500 Internal Server Error	HTTP 403/404	HTTP 500 msg="Internal server error"	FAIL
NR-B01	Limit value (min / max / just outside)	PATCH /notifications/read/0 (id=0) Note: SERVER ERROR 500	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	FAIL
NR-B02	Limit value (min / max / just outside)	PATCH /notifications/read/-1 (id=-1) Note: SERVER ERROR 500	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	FAIL
NR-B03	Limit value (min / max / just outside)	PATCH /notifications/read/abc (id=abc) Note: SERVER ERROR 500	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	FAIL
NR-B04	Limit value (min / max / just outside)	PATCH /notifications/read/999999999 (id=999999999 (missing)) Note: SERVER ERROR 500	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	FAIL
NR-B05	Limit value (min / max / just outside)	PATCH /notifications/read/1.5 (id=1.5) Note: SERVER ERROR 500	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	FAIL
NR-B06	Limit value (min / max / just outside)	PATCH /notifications/read/99999999999999999 (id beyond int range) Note: SERVER ERROR 500	HTTP 400/404 (never 500)	HTTP 500 msg="Internal server error"	FAIL
NRA-P01	Normal use (should work)	PATCH /notifications/readAll returns {message, count}	HTTP 200 + count number	HTTP 200 body={"message":"All notifications marked as read","count":4}	PASS
NRA-P02	Normal use (should work)	After readAll the non-MESSAGE unread count is 0	0	0	PASS
NRA-E01	Unusual situation	readAll again → count 0 (idempotent)	HTTP 200, count 0	HTTP 200 count=0	PASS
NRA-P03	Normal use (should work)	PATCH /notifications/messages/readAll marks MESSAGE notifications read	HTTP 200	HTTP 200 body={"message":"All notifications marked as read","count":2}	PASS
NRA-P04	Normal use (should work)	After messages/readAll, /notifications/chat is empty	0 unread MESSAGE	n=0	PASS
NRM-N01	Wrong input (should be refused)	PATCH /notifications/messages/readAll/undefined (chatroomId=undefined (FE bug value → NaN))	HTTP 400/404, never 500	HTTP 200 msg="All Messages notifications marked as read for 27 NaN"	FAIL
NRM-B01	Limit value (min / max / just outside)	PATCH /notifications/messages/readAll/0 (chatroomId=0)	HTTP 200/400/404, never 500	HTTP 200 msg="All Messages notifications marked as read for 27 0"	PASS
NRM-B02	Limit value (min / max / just outside)	PATCH /notifications/messages/readAll/abc (chatroomId=abc)	HTTP 400/404, never 500	HTTP 200 msg="All Messages notifications marked as read for 27 NaN"	FAIL
NRM-B03	Limit value (min / max / just outside)	PATCH /notifications/messages/readAll/999999999 (chatroomId missing)	HTTP 200/404, never 500	HTTP 200 msg="All Messages notifications marked as read for 27 999999999"	PASS
NRM-B04	Limit value (min / max / just outside)	PATCH /notifications/messages/readAll/-1 (chatroomId=-1)	HTTP 200/400/404, never 500	HTTP 200 msg="All Messages notifications marked as read for 27 -1"	PASS
ND-N01	Wrong input (should be refused)	DELETE /notifications/:id does not exist (404)	HTTP 404/405	HTTP 404 / 404	PASS
NS-P01	Normal use (should work)	Customer work-order notifications carry ?fromCustomerBoard=true	all WO urls have fromCustomerBoard=true	n=10 without flag=0	PASS
NS-P02	Normal use (should work)	Customer can list message notifications (shape ok)	HTTP 200 shape	HTTP 200 keys=data,nextCursor,unreadNotificationCount	PASS
NT-N01	Wrong input (should be refused)	PATCH /users/deviceToken with empty body is rejected Note: accepts empty body (clears token?) — informational	HTTP 400	HTTP 200 msg="SUCCESS"	FAIL
NT-N02	Wrong input (should be refused)	PATCH /users/deviceToken without token → 401	HTTP 401	HTTP 401	PASS

Chat & inbox rules (§4) (61 pass / 6 fail)

ID	Kind	What we checked	What should happen	What happened	Resu
CH-N01	Wrong input (should be refused)	GET /inbox without token → 401	HTTP 401	HTTP 401	PAS
CH-N02	Wrong input (should be refused)	GET /chats/1 without token → 401	HTTP 401	HTTP 401	PAS
CH-N03	Wrong input (should be refused)	GET /chats/users/1/1 without token → 401	HTTP 401	HTTP 401	PAS
CH-N04	Wrong input (should be refused)	POST /inbox/1/message without token → 401	HTTP 401	HTTP 401	PAS
CH-N05	Wrong input (should be refused)	PATCH /inbox/readAll without token → 401	HTTP 401	HTTP 401	PAS
CU-E01	Limit value (min / max / just outside)	GET /chats/users/abc/xyz (non-integer params) → empty list, never 500	HTTP 200 + [] (or 400)	HTTP 400 body={"statusCode":400,"message":"Invalid companyId provided"}	PAS
CU-E02	Limit value (min / max / just outside)	GET /chats/users/0/0 (zeros) → empty list, never 500	HTTP 200 + [] (or 400)	HTTP 200 body={"data":[],"message":"SUCCESS"}	PAS
CU-E03	Limit value (min / max / just outside)	GET /chats/users/-1/-1 (negatives) → empty list, never 500	HTTP 200 + [] (or 400)	HTTP 200 body={"data":[],"message":"SUCCESS"}	PAS
CU-E04	Limit value (min / max / just outside)	GET /chats/users/999999999/999999999 (missing ids) → empty list, never 500	HTTP 200 + [] (or 400)	HTTP 200 body={"data":[],"message":"SUCCESS"}	PAS
IN-P01	Normal use (should work)	GET /inbox returns {data, nextCursor, totalUnreadCount, unreadConversationCount}	shape + ≤20 rows	HTTP 200 keys=data,nextCursor,totalUnreadCount,unreadConversationCount,message n=15	PAS
IN-P02	Normal use (should work)	Conversation row has conversationId/chatRoomId/workOrder/participant/latestMessage/unreadCount	fields present	keys=conversationId,chatRoomId,workOrder,engineerId,isWorkOrderChat,isDisabled,disabledMessage,company,participant,participantCount,latestMessage,latestMessageAt,unreadCount	PAS
IN-P03	Normal use (should work)	Preview text starts with "You: " or "<FirstName>: "	prefix	Demo: hi	PAS
IN-P04	Normal use	Only rooms with at least one message are listed	every row has latestMessage	rows without latestMessage=0	PAS

	(should work)				
IN-P05	Unusual situation	totalUnreadCount equals the sum of row unread counts (page 1 only when no more pages)	0 (if single page)	total=0 nextCursor=null	PAS
IN-B-0	Limit value (min / max / just outside)	GET /inbox limit=0 Note: accepted limit=0, returned 15	HTTP 200/400, rows ≤ min(limit,50)	HTTP 200 rows=15	PAS
IN-B-1	Limit value (min / max / just outside)	GET /inbox limit=1	HTTP 200, rows ≤ min(limit,50)	HTTP 200 rows=1	PAS
IN-B-20	Limit value (min / max / just outside)	GET /inbox limit=20 (default)	HTTP 200, rows ≤ min(limit,50)	HTTP 200 rows=15	PAS
IN-B-21	Limit value (min / max / just outside)	GET /inbox limit=21	HTTP 200/400, rows ≤ min(limit,50)	HTTP 200 rows=15	PAS
IN-B-50	Limit value (min / max / just outside)	GET /inbox limit=50 (max)	HTTP 200, rows ≤ min(limit,50)	HTTP 200 rows=15	PAS
IN-B-51	Limit value (min / max / just outside)	GET /inbox limit=51 (above max → capped or 400)	HTTP 200/400, rows ≤ min(limit,50)	HTTP 200 rows=15	PAS
IN-B--1	Limit value (min / max / just outside)	GET /inbox limit=-1 Note: accepted limit=-1, returned 1	HTTP 200/400, rows ≤ min(limit,50)	HTTP 200 rows=1	PAS
IN-B-abc	Limit value (min / max / just outside)	GET /inbox limit=abc Note: accepted limit=abc, returned 15	HTTP 200/400, rows ≤ min(limit,50)	HTTP 200 rows=15	PAS
IN-C-malformedcursor	Unusual situation	GET /inbox with malformed cursor falls back to page 1	HTTP 200 + rows	HTTP 200 rows=5	PAS
IN-C-symbols	Unusual situation	GET /inbox with symbols falls back to page 1	HTTP 200 + rows	HTTP 200 rows=5	PAS
IN-C-base64withbadid	Unusual situation	GET /inbox with base64 with bad id falls back to page 1	HTTP 200 + rows	HTTP 200 rows=5	PAS
CHH-P01	Normal use (should work)	GET /chats/:roomId returns {messages, nextCursor, hasMore}, default page 10	shape + ≤10 messages ascending	HTTP 200 keys=messages,nextCursor,hasMore n=1	PAS
CHH-P02	Normal use (should work)	Messages are in ascending id order	ascending	8387	PAS

CHH-B-0	Limit value (min / max / just outside)	GET /chats/:roomId limit=0	HTTP 200/400, n ≤ min(limit,100)	HTTP 200 n=1	PAS
CHH-B-1	Limit value (min / max / just outside)	GET /chats/:roomId limit=1	HTTP 200, n ≤ min(limit,100)	HTTP 200 n=1	PAS
CHH-B-100	Limit value (min / max / just outside)	GET /chats/:roomId limit=100 (max)	HTTP 200, n ≤ min(limit,100)	HTTP 200 n=1	PAS
CHH-B-101	Limit value (min / max / just outside)	GET /chats/:roomId limit=101 (above max)	HTTP 200/400, n ≤ min(limit,100)	HTTP 200 n=1	PAS
CHH-B--3	Limit value (min / max / just outside)	GET /chats/:roomId limit=-3	HTTP 200/400, n ≤ min(limit,100)	HTTP 200 n=1	PAS
CHH-B-x	Limit value (min / max / just outside)	GET /chats/:roomId limit=x	HTTP 200/400, n ≤ min(limit,100)	HTTP 200 n=1	PAS
CHH-B-abc	Limit value (min / max / just outside)	GET /chats/abc (non-numeric room) Note: SERVER ERROR 500	HTTP 400/404, never 500	HTTP 500	FAI
CHH-N01	Wrong input (should be refused)	GET /chats/999999999 (missing room)	HTTP 403/404	HTTP 403	PAS
CHH-N02	Wrong input (should be refused)	Non-participant (Customer) cannot read the room	HTTP 403	HTTP 403	PAS
CHH-N03	Wrong input (should be refused)	Non-participant (other engineer) cannot read the room	HTTP 403	HTTP 403	PAS
CHH-P04	Normal use (should work)	Freelance company owner can read the member room	HTTP 200	HTTP 200	PAS
IT-P01	Normal use (should work)	GET /inbox/:id returns the thread with participants	HTTP 200 + messages + participants	HTTP 200 keys=conversation,messages,previousCursor,nextCursor	PAS
IT-N01	Wrong input (should	GET /inbox/:id as non-participant → 403	HTTP 403	HTTP 403	PAS

	be refused)				
IT-B-abc	Limit value (min / max / just outside)	GET /inbox/abc (id=abc) Note: SERVER ERROR 500	HTTP 400/403/404, never 500	HTTP 500	FAI
IT-B-0	Limit value (min / max / just outside)	GET /inbox/0 (id=0)	HTTP 400/403/404, never 500	HTTP 403	PAS
IT-B--1	Limit value (min / max / just outside)	GET /inbox/-1 (id=-1)	HTTP 400/403/404, never 500	HTTP 403	PAS
IT-B-999999999	Limit value (min / max / just outside)	GET /inbox/999999999 (missing id)	HTTP 400/403/404, never 500	HTTP 403	PAS
SM-N01	Wrong input (should be refused)	Send rejected: empty body	HTTP 400	HTTP 400 msg="message should not be empty | message must be a string"	PAS
SM-N02	Wrong input (should be refused)	Send rejected: empty message	HTTP 400	HTTP 400 msg="message should not be empty"	PAS
SM-N03	Wrong input (should be refused)	Send rejected: message as number	HTTP 400	HTTP 400 msg="message must be a string"	PAS
SM-N04	Wrong input (should be refused)	Send rejected: message null	HTTP 400	HTTP 400 msg="message should not be empty | message must be a string"	PAS
SM-B01	Limit value (min / max / just outside)	Send rejected: 21 digitalAssetIds (max 20)	HTTP 400	HTTP 400 msg="digitalAssetIds must contain no more than 20 elements"	PAS
SM-B02	Limit value (min / max / just outside)	Send rejected: clientMessageId 65 chars (max 64)	HTTP 400	HTTP 400 msg="clientMessageId must be shorter than or equal to 64 characters"	PAS
SM-N05	Wrong input (should be refused)	Send rejected: digitalAssetIds not an array	HTTP 400	HTTP 400 msg="each value in digitalAssetIds must be an integer number | digitalAssetIds must contain no more than 20 elements | digitalAssetIds must be an array"	PAS
SM-N06	Wrong input (should be refused)	Send rejected: replyToId non-numeric	HTTP 400/404	HTTP 400 msg="replyToId must be an integer number"	PAS

SM-P01	Normal use (should work)	Valid message with 64-char clientMessageId is sent	HTTP 200/201 + sentMessage	HTTP 201 id=9135 keys=sentMessage,conversation	PAS
SM-E01	Unusual situation	Resend with the same clientMessageId returns the original message (no duplicate)	same id 9135	HTTP 201 id=9135 isReplay=undefined	PAS
SM-B03	Limit value (min / max / just outside)	280-char message accepted; inbox preview clipped at 200	HTTP 201; preview length ≤ 200+prefix	HTTP 201	PAS
SM-B04	Limit value (min / max / just outside)	Inbox preview of the long message is clipped (≤ ~205 chars)	≤ 205	preview length=205	PAS
SM-E02	Unusual situation	Script tag in a message is stored/escaped (XSS hygiene) Note: RAW <script> STORED — FE must sanitise on render; security note	<script> removed or escaped	HTTP 201 stored="[CL-04 1014101612] <script>alert(1)</script> bold"	FAI
SM-N07	Wrong input (should be refused)	Non-participant cannot send (403)	HTTP 403	HTTP 403	PAS
SM-N08	Wrong input (should be refused)	Send to a missing room → 404	HTTP 404	HTTP 404	PAS
SM-P02	Normal use (should work)	Reader's inbox shows unread count > 0 for the room after the sends	unreadCount ≥ 1	unreadCount=3 total=3	PAS
RD-P01	Normal use (should work)	PATCH /inbox/:id/read returns {unreadConversationCount}	HTTP 200 + unreadConversationCount	HTTP 200 body={"data":{"unreadConversationCount":0},"message":"SUCCESS"}	PAS
RD-P02	Normal use (should work)	After read the room unread count is 0	0	0	PAS
RD-B01	Limit value (min / max / just outside)	PATCH /inbox/abc/read Note: SERVER ERROR 500	HTTP 400/404, never 500	HTTP 500	FAI
RA-P01	Normal use (should work)	PATCH /inbox/readAll returns {conversationsCleared, totalUnreadCount, unreadConversationCount}	HTTP 200 + 3 counters	HTTP 200 keys=conversationsCleared,totalUnreadCount,unreadConversationCount	PAS
RA-E01	Unusual situation	readAll twice → conversationsCleared 0 and totalUnreadCount 0	0 / 0	cleared=0 total=0	PAS
AT-N01	Wrong input (should be refused)	POST /chats/files without files is rejected (not 500) Note: accepted an upload with no files (GAP: no limits)	HTTP 400	HTTP 201 msg="{"fileIds":[]}"	FAI
AT-E01	Unusual situation	Uploading an .exe as a chat file Note: CONFIRMED GAP: executable accepted as chat attachment	HTTP 400 (type rejected) — GAP if accepted	HTTP 201 body={"data":[3829],"message":"SUCCESS"}	FAI

Roles & permissions (§6) (113 pass / 3 fail)

ID	Kind	What we checked	What should happen	What happened	Result
PM-Admin-ADMIN_NEW_USERS_MANAGEMENT	Normal use (should work)	Admin has ADMIN_NEW_USERS_MANAGEMENT = RWUD	RWUD	RWUD	PASS
PM-Admin-MANAGE	Normal use (should work)	Admin has MANAGE = RWUD	RWUD	RWUD	PASS
PM-Admin-COMPANIES	Normal use (should work)	Admin has COMPANIES = RWUD	RWUD	RWUD	PASS
PM-Admin-USERS	Normal use (should work)	Admin has USERS = RWUD	RWUD	RWUD	PASS
PM-Admin-WORK_EXPERIENCE	Normal use (should work)	Admin has WORK_EXPERIENCE = RWUD	RWUD	RWUD	PASS
PM-Admin-QUALIFICATIONS	Normal use (should work)	Admin has QUALIFICATIONS = RWUD	RWUD	RWUD	PASS
PM-Admin-no-WORK_ORDERS	Wrong input (should be refused)	Admin has NO WORK_ORDERS row	none	none	PASS
PM-Admin-no-JOBS	Wrong input (should be refused)	Admin has NO JOBS row	none	none	PASS
PM-Admin-dup	Unusual situation	Admin: no duplicate module rows	unique modules	rows=6 unique=6	PASS
PM-TechDomain User-WORK_ORDERS	Normal use (should work)	TechDomain User has WORK_ORDERS = RWUD	RWUD	RWUD	PASS
PM-TechDomain User-COMPANIES	Normal use (should work)	TechDomain User has COMPANIES = RWUD	RWUD	RWUD	PASS
PM-TechDomain User-USER_PERMISSIONS	Normal use (should work)	TechDomain User has USER_PERMISSIONS = RWUD	RWUD	RWUD	PASS
PM-TechDomain User-JOBS	Normal use (should work)	TechDomain User has JOBS = RWUD	RWUD	RWUD	PASS
PM-TechDomain User-INVOICE	Normal use (should work)	TechDomain User has INVOICE = RWUD	RWUD	RWUD	PASS
PM-TechDomain User-TASK_BOARD	Normal use (should work)	TechDomain User has TASK_BOARD = RWUD	RWUD	RWUD	PASS
PM-TechDomain User-CUSTOMER_WORK_ORDERS	Normal use (should work)	TechDomain User has CUSTOMER_WORK_ORDERS = RWUD	RWUD	RWUD	PASS
PM-TechDomain User-ACTION_APPROVE_WORK_ORDER	Normal use (should work)	TechDomain User has ACTION_APPROVE_WORK_ORDER = RWUD	RWUD	RWUD	PASS
PM-TechDomain User-no-ADMIN_NEW_USERS_MANAGEMENT	Wrong input (should be refused)	TechDomain User has NO ADMIN_NEW_USERS_MANAGEMENT row	none	none	PASS
PM-TechDomain User-no-MANAGE	Wrong input (should be refused)	TechDomain User has NO MANAGE row	none	none	PASS
PM-TechDomain User-no-CLIENT_WORK_ORDERS	Wrong input (should be refused)	TechDomain User has NO CLIENT_WORK_ORDERS row	none	none	PASS
PM-BUG-WORK_FORCE_ENGINEERS	Unusual situation	Client owner WORK_FORCE_ENGINEERS (checklist [BUG?]: missing for self-signup owners / R-only coverage map)	RWUD ideally	RWUD	PASS
PM-BUG-WORK_FORCE_TALENT_POOL	Unusual situation	Client owner WORK_FORCE_TALENT_POOL (checklist [BUG?]: missing for self-signup owners / R-only coverage map)	RWUD ideally	RWUD	PASS
PM-BUG-WORK_FORCE_COVERAGE_MAP	Unusual situation	Client owner WORK_FORCE_COVERAGE_MAP (checklist [BUG?]: missing for self-signup owners / R-only coverage map)	RWUD ideally	RWUD	PASS
PM-TechDomain User-dup	Unusual situation	TechDomain User: no duplicate module rows	unique modules	rows=28 unique=28	PASS
PM-Individual Engineer-JOBS	Normal use (should work)	Individual Engineer has JOBS = /RWU/	/RWU/	RWU-	PASS
PM-Individual Engineer-USERS	Normal use (should work)	Individual Engineer has USERS = /R/	/R/	R---	PASS
PM-Individual Engineer-no-WORK_ORDERS	Wrong input (should be refused)	Individual Engineer has NO WORK_ORDERS row	none	none	PASS
PM-Individual Engineer-no-COMPANIES	Wrong input (should be refused)	Individual Engineer has NO COMPANIES row	none	none	PASS
PM-Individual Engineer-no-INVOICE	Wrong input (should be refused)	Individual Engineer has NO INVOICE row	none	none	PASS

PM-Individual Engineer-no-ADMIN_NEW_USERS_MANAGEMENT	Wrong input (should be refused)	Individual Engineer has NO ADMIN_NEW_USERS_MANAGEMENT row	none	none	**PASS**
PM-Individual Engineer-dup	Unusual situation	Individual Engineer: no duplicate module rows	unique modules	rows=2 unique=2	**PASS**
PM-Vendor-JOBS	Normal use (should work)	Vendor has JOBS = /RWU/	/RWU/	RWU-	**PASS**
PM-Vendor-USERS	Normal use (should work)	Vendor has USERS = /R/	/R/	R---	**PASS**
PM-Vendor-no-WORK_ORDERS	Wrong input (should be refused)	Vendor has NO WORK_ORDERS row	none	none	**PASS**
PM-Vendor-no-ADMIN_NEW_USERS_MANAGEMENT	Wrong input (should be refused)	Vendor has NO ADMIN_NEW_USERS_MANAGEMENT row	none	none	**PASS**
PM-Vendor-dup	Unusual situation	Vendor: no duplicate module rows	unique modules	rows=3 unique=3	**PASS**
PM-Vendor Engineer-JOBS	Normal use (should work)	Vendor Engineer has JOBS = /RWU/	/RWU/	RWU-	**PASS**
PM-Vendor Engineer-USERS	Normal use (should work)	Vendor Engineer has USERS = /R/	/R/	R---	**PASS**
PM-Vendor Engineer-no-WORK_ORDERS	Wrong input (should be refused)	Vendor Engineer has NO WORK_ORDERS row	none	none	**PASS**
PM-Vendor Engineer-no-FREELANCE_COMPANY_WORKERS	Wrong input (should be refused)	Vendor Engineer has NO FREELANCE_COMPANY_WORKERS row	none	none	**PASS**
PM-Vendor Engineer-no-ADMIN_NEW_USERS_MANAGEMENT	Wrong input (should be refused)	Vendor Engineer has NO ADMIN_NEW_USERS_MANAGEMENT row	none	none	**PASS**
PM-Vendor Engineer-dup	Unusual situation	Vendor Engineer: no duplicate module rows	unique modules	rows=2 unique=2	**PASS**
PM-Customer-CLIENT_WORK_ORDERS	Normal use (should work)	Customer has CLIENT_WORK_ORDERS = RWUD	RWUD	RWUD	**PASS**
PM-Customer-INVOICE	Normal use (should work)	Customer has INVOICE = /R/	/R/	R---	**PASS**
PM-Customer-CUSTOMER_INVOICE	Normal use (should work)	Customer has CUSTOMER_INVOICE = /R/	/R/	R---	**PASS**
PM-Customer-ACTION_SEND_CUSTOMER_INVOICE	Normal use (should work)	Customer has ACTION_SEND_CUSTOMER_INVOICE = /RW/	/RW/	RW--	**PASS**
PM-Customer-USERS	Normal use (should work)	Customer has USERS = RWUD	RWUD	RWUD	**PASS**
PM-Customer-no-ACTION_VERIFY_CUSTOMER_INVOICE	Wrong input (should be refused)	Customer has NO ACTION_VERIFY_CUSTOMER_INVOICE row	none	none	**PASS**
PM-Customer-no-WORK_ORDERS	Wrong input (should be refused)	Customer has NO WORK_ORDERS row	none	none	**PASS**
PM-Customer-no-ADMIN_NEW_USERS_MANAGEMENT	Wrong input (should be refused)	Customer has NO ADMIN_NEW_USERS_MANAGEMENT row	none	none	**PASS**
PM-Customer-no-ACTION_PAID_CUSTOMER_INVOICE	Wrong input (should be refused)	Customer has NO ACTION_PAID_CUSTOMER_INVOICE row	none	none	**PASS**
PM-Customer-dup	Unusual situation	Customer: no duplicate module rows	unique modules	rows=10 unique=10	**PASS**
PM-Customer User-CLIENT_WORK_ORDERS	Normal use (should work)	Customer User has CLIENT_WORK_ORDERS = RWUD	RWUD	RWUD	**PASS**
PM-Customer User-ACTION_SEND_CUSTOMER_INVOICE	Normal use (should work)	Customer User has ACTION_SEND_CUSTOMER_INVOICE = /RW/	/RW/	RW--	**PASS**
PM-Customer User-no-ACTION_VERIFY_CUSTOMER_INVOICE	Wrong input (should be refused)	Customer User has NO ACTION_VERIFY_CUSTOMER_INVOICE row	none	none	**PASS**
PM-Customer User-no-CUSTOMER_USERS	Wrong input (should be refused)	Customer User has NO CUSTOMER_USERS row	none	none	**PASS**
PM-Customer User-no-WORK_ORDERS	Wrong input (should be refused)	Customer User has NO WORK_ORDERS row	none	none	**PASS**
PM-Customer User-dup	Unusual situation	Customer User: no duplicate module rows	unique modules	rows=9 unique=9	**PASS**
XR-01	Wrong input (should be refused)	Individual Engineer → POST /admin/inviteEngineer is refused (ADMIN only)	HTTP 401/403 (404 acceptable)	HTTP 403 msg="Forbidden resource"	**PASS**
XR-02	Wrong input (should be refused)	Customer → POST /admin/inviteEngineer is refused (ADMIN only)	HTTP 401/403 (404 acceptable)	HTTP 403 msg="Forbidden resource"	**PASS**
XR-03	Wrong input (should be refused)	Individual Engineer → GET /clients is refused (CLIENT_CUSTOMER read)	HTTP 401/403 (404 acceptable)	HTTP 403 msg="Forbidden resource"	**PASS**

XR-04	Wrong input (should be refused)	Customer → GET /teams is refused (COMPANIES read) Note: ACCESS ALLOWED — permission gap	HTTP 401/403 (404 acceptable)	HTTP 200 msg="SUCCESS"	**FAIL**
XR-05	Wrong input (should be refused)	Individual Engineer → GET /projects is refused (CLIENT_PROJECT read)	HTTP 401/403 (404 acceptable)	HTTP 403 msg="Forbidden resource"	**PASS**
XR-06	Wrong input (should be refused)	Customer → POST /offers/route is refused (WORK_ORDERS write) Note: validation ran before the permission check (informational)	HTTP 401/403 (404 acceptable)	HTTP 400 msg="Select at least one engineer to route."	**FAIL**
XR-07	Wrong input (should be refused)	TechDomain User → GET /admin/review/engineers is refused (ADMIN only)	HTTP 401/403 (404 acceptable)	HTTP 404 msg="Cannot GET /api/admin/review/engineers"	**PASS**
XR-08	Wrong input (should be refused)	Individual Engineer → GET /admin/approveProfile/999999999 is refused (ADMIN only)	HTTP 401/403 (404 acceptable)	HTTP 403 msg="Forbidden resource"	**PASS**
XR-09	Wrong input (should be refused)	Customer User → POST /customerUsers/createUser is refused (CUSTOMER_USERS (sub-user has none))	HTTP 401/403 (404 acceptable)	HTTP 403 msg="Only the customer account owner can manage users"	**PASS**
XR-10	Wrong input (should be refused)	Vendor Engineer → POST /freelanceCompany/createEngineer is refused (company owner only)	HTTP 401/403 (404 acceptable)	HTTP 403 msg="Only freelancer company owners can create workers"	**PASS**
XR-11	Wrong input (should be refused)	Individual Engineer → GET /customerUsers is refused (CUSTOMER only)	HTTP 401/403 (404 acceptable)	HTTP 404 msg="Cannot GET /api/customerUsers"	**PASS**
XR-12	Wrong input (should be refused)	Customer → GET /users/engineer/submitProfile is refused (FREELANCER only) Note: validation ran before the permission check (informational)	HTTP 401/403 (404 acceptable)	HTTP 400 msg="Complete these required fields before submitting your profile: Nationality, ID document, ABN or Foreign Contractor declaration, Hourly, Half Day & Full Day rates, Resume"	**FAIL**
UI-Admin-land	Normal use (should work)	Admin lands on /\/admin\/review\/engineers/	/\/admin\/review\/engineers/	https://dev.techdomain.com.au/admin/review/engineers	**PASS**
UI-Admin-has-Accounts	Normal use (should work)	Admin sidebar shows "Accounts"	Accounts	Manage | Accounts | View | View | View | View | File 1 | File 2 | View | View | View | View | View	**PASS**
UI-Admin-has-Manage	Normal use (should work)	Admin sidebar shows "Manage"	Manage	Manage | Accounts | View | View | View | View | File 1 | File 2 | View | View | View | View | View	**PASS**
UI-Admin-no-Work Board	Wrong input (should be refused)	Admin sidebar hides "Work Board"	hidden	hidden	**PASS**
UI-Admin-no-Job	Wrong input (should be refused)	Admin sidebar hides "Job"	hidden	hidden	**PASS**
UI-Admin-foreign	Wrong input (should be refused)	Admin opening /workBoard/workOrders gets Unauthorized / redirected	Unauthorized text or redirect away	https://dev.techdomain.com.au/workBoard/workOrders text=Manage Accounts 0 5 Unauthorized	**PASS**
UI-TechDomain User-land	Normal use (should work)	TechDomain User lands on /\/workBoard\/workOrders/	/\/workBoard\/workOrders/	https://dev.techdomain.com.au/workBoard/workOrders	**PASS**
UI-TechDomain User-has-Dashboard	Normal use (should work)	TechDomain User sidebar shows "Dashboard"	Dashboard	Dashboard | Work Board | Task Board | Work Force | Invoices | PO-0000035935/90 | PO-0000035935/90 | PO99988800	**PASS**
UI-TechDomain User-has-Work Board	Normal use (should work)	TechDomain User sidebar shows "Work Board"	Work Board	Dashboard | Work Board | Task Board | Work Force | Invoices | PO-0000035935/90 | PO-0000035935/90 | PO99988800	**PASS**
UI-TechDomain User-has-Task Board	Normal use (should work)	TechDomain User sidebar shows "Task Board"	Task Board	Dashboard | Work Board | Task Board | Work Force | Invoices | PO-0000035935/90 | PO-0000035935/90 | PO99988800	**PASS**
UI-TechDomain User-has-Invoices	Normal use (should work)	TechDomain User sidebar shows "Invoices"	Invoices	Dashboard | Work Board | Task Board | Work Force | Invoices | PO-0000035935/90 | PO-0000035935/90 | PO99988800	**PASS**
UI-TechDomain User-no-Accounts	Wrong input (should be refused)	TechDomain User sidebar hides "Accounts"	hidden	hidden	**PASS**
UI-TechDomain User-no-Manage	Wrong input (should be refused)	TechDomain User sidebar hides "Manage"	hidden	hidden	**PASS**
UI-TechDomain User-foreign	Wrong input (should be refused)	TechDomain User opening /admin/review/engineers gets Unauthorized / redirected	Unauthorized text or redirect away	https://dev.techdomain.com.au/admin/review/engineers text=Dashboard Work Board Task Board Work Force Invoices 14 0 Una	**PASS**
UI-Individual Engineer-land	Normal use (should work)	Individual Engineer lands on /\/jobs/	/\/jobs/	https://dev.techdomain.com.au/jobs	**PASS**
UI-Individual Engineer-has-Analytics	Normal use (should work)	Individual Engineer sidebar shows "Analytics"	Analytics	Analytics | Job	**PASS**
UI-Individual Engineer-has-Job	Normal use (should work)	Individual Engineer sidebar shows "Job"	Job	Analytics | Job	**PASS**
UI-Individual Engineer-no-Workers	Wrong input (should be refused)	Individual Engineer sidebar hides "Workers"	hidden	hidden	**PASS**
UI-Individual Engineer-no-Work Board	Wrong input (should be refused)	Individual Engineer sidebar hides "Work Board"	hidden	hidden	**PASS**
UI-Individual Engineer-no-Invoices	Wrong input (should be refused)	Individual Engineer sidebar hides "Invoices"	hidden	hidden	**PASS**
UI-Individual Engineer-foreign	Wrong input (should be refused)	Individual Engineer opening /admin/review/engineers gets Unauthorized / redirected	Unauthorized text or redirect away	https://dev.techdomain.com.au/admin/review/engineers text=Analytics Job 0 11 Unauthorized	**PASS**

UI-Vendor-land	Normal use (should work)	Vendor lands on /\/jobs/	/\/jobs/	https://dev.techdomain.com.au/jobs	**PASS**
UI-Vendor-has-Analytics	Normal use (should work)	Vendor sidebar shows "Analytics"	Analytics	Analytics | Job | Workers	**PASS**
UI-Vendor-has-Job	Normal use (should work)	Vendor sidebar shows "Job"	Job	Analytics | Job | Workers	**PASS**
UI-Vendor-has-Workers	Normal use (should work)	Vendor sidebar shows "Workers"	Workers	Analytics | Job | Workers	**PASS**
UI-Vendor-no-Work Board	Wrong input (should be refused)	Vendor sidebar hides "Work Board"	hidden	hidden	**PASS**
UI-Vendor-foreign	Wrong input (should be refused)	Vendor opening /admin/review/engineers gets Unauthorized / redirected	Unauthorized text or redirect away	https://dev.techdomain.com.au/admin/review/engineers text=Analytics Job Workers 0 89 Unauthorized	**PASS**
UI-Vendor Engineer-land	Normal use (should work)	Vendor Engineer lands on /\/jobs/	/\/jobs/	https://dev.techdomain.com.au/jobs	**PASS**
UI-Vendor Engineer-has-Analytics	Normal use (should work)	Vendor Engineer sidebar shows "Analytics"	Analytics	Analytics | Job	**PASS**
UI-Vendor Engineer-has-Job	Normal use (should work)	Vendor Engineer sidebar shows "Job"	Job	Analytics | Job	**PASS**
UI-Vendor Engineer-no-Workers	Wrong input (should be refused)	Vendor Engineer sidebar hides "Workers"	hidden	hidden	**PASS**
UI-Vendor Engineer-no-Work Board	Wrong input (should be refused)	Vendor Engineer sidebar hides "Work Board"	hidden	hidden	**PASS**
UI-Vendor Engineer-foreign	Wrong input (should be refused)	Vendor Engineer opening /admin/review/engineers gets Unauthorized / redirected	Unauthorized text or redirect away	https://dev.techdomain.com.au/admin/review/engineers text=Analytics Job 0 0 Unauthorized	**PASS**
UI-Customer-land	Normal use (should work)	Customer lands on /\/workOrders/	/\/workOrders/	https://dev.techdomain.com.au/workOrders	**PASS**
UI-Customer-has-Dashboard	Normal use (should work)	Customer sidebar shows "Dashboard"	Dashboard	Dashboard | Work Orders | Work Force | Invoices	**PASS**
UI-Customer-has-Work Orders	Normal use (should work)	Customer sidebar shows "Work Orders"	Work Orders	Dashboard | Work Orders | Work Force | Invoices	**PASS**
UI-Customer-has-Invoices	Normal use (should work)	Customer sidebar shows "Invoices"	Invoices	Dashboard | Work Orders | Work Force | Invoices	**PASS**
UI-Customer-no-Task Board	Wrong input (should be refused)	Customer sidebar hides "Task Board"	hidden	hidden	**PASS**
UI-Customer-no-Accounts	Wrong input (should be refused)	Customer sidebar hides "Accounts"	hidden	hidden	**PASS**
UI-Customer-foreign	Wrong input (should be refused)	Customer opening /admin/review/engineers gets Unauthorized / redirected	Unauthorized text or redirect away	https://dev.techdomain.com.au/admin/review/engineers text=Dashboard Work Orders Work Force Invoices 2 2 Unauthorized	**PASS**
UI-Customer User-land	Normal use (should work)	Customer User lands on /\/workOrders/	/\/workOrders/	https://dev.techdomain.com.au/workOrders	**PASS**
UI-Customer User-has-Dashboard	Normal use (should work)	Customer User sidebar shows "Dashboard"	Dashboard	Dashboard | Work Orders | Work Force | Invoices	**PASS**
UI-Customer User-has-Work Orders	Normal use (should work)	Customer User sidebar shows "Work Orders"	Work Orders	Dashboard | Work Orders | Work Force | Invoices	**PASS**
UI-Customer User-no-Task Board	Wrong input (should be refused)	Customer User sidebar hides "Task Board"	hidden	hidden	**PASS**
UI-Customer User-no-Accounts	Wrong input (should be refused)	Customer User sidebar hides "Accounts"	hidden	hidden	**PASS**
UI-Customer User-foreign	Wrong input (should be refused)	Customer User opening /admin/review/engineers gets Unauthorized / redirected	Unauthorized text or redirect away	https://dev.techdomain.com.au/admin/review/engineers text=Dashboard Work Orders Work Force Invoices 1 49 Unauthorized	**PASS**

Onboarding — Complete Your Profile wizard (27 pass / 5 fail)

ID	Kind	What we checked	What should happen	What happened	Result
OB-P01	Normal use (should work)	PENDING engineer: GET users/me works and status is PENDING	HTTP 200, status PENDING	HTTP 200 status=PENDING	PASS
OB-P02	Normal use (should work)	GET qualifications (qualification types) works	HTTP 200 + list	HTTP 200 n=undefined	PASS
OB-N01	Wrong input (should be refused)	Submit with an empty profile is rejected	HTTP 400 + reason	HTTP 400 msg="Complete these required fields before submitting your profile: Nationality, ID document, Address, Country, City, State, ABN or Foreign Contractor declaration, Hourly, Half Day & Full Day rates, Resume"	PASS
OB-B01	Limit value (min / max / just outside)	About me 1000 chars accepted	HTTP 200	HTTP 200 msg="SUCCESS"	PASS
OB-B02	Limit value (min / max / just outside)	About me 1001 chars rejected by the server Note: SERVER ACCEPTS >1000 chars — only the UI enforces the limit	HTTP 400	HTTP 200 msg="SUCCESS"	FAIL
OB-B03	Limit value (min / max / just outside)	About me cleared back to empty (revert)	HTTP 200	HTTP 200	PASS
OB-E01	Unusual situation	HTML with onerror in About me is sanitised Note: RAW HTML STORED — FE must sanitise on render (security note)	onerror stripped	HTTP 200 stored="<img src=x onerror=alert(1)>"	FAIL
OB-N02	Wrong input (should be refused)	PATCH users/address with empty body is rejected	HTTP 400	HTTP 200 msg="SUCCESS"	FAIL
OB-N03	Wrong input (should be refused)	PATCH users/address with blank country/city/state is rejected Note: SERVER ACCEPTED blank address fields (UI-only validation) — reverted	HTTP 400	HTTP 200 msg="SUCCESS"	FAIL
OB-B04	Limit value (min / max / just outside)	Billing rates of 0 without a home address	HTTP 400 (or 200 saving zeros)	HTTP 400 msg="billingRates.0.paymentType must be one of the following values: HOURLY, FULL_DAY, HALF_DAY, FIXED, FTH, CUSTOM_HOURS, PER_DEVICE, BLENDED | billingRates.0.currency must be one of the following values: AED, AFN, ALL, AMD, ANG, AOA, ARS, AUD, AWG, AZN, BAM, BBD, BDT, BGN, BHD, BIF, BMD, BND, BOB, BRL, BSD, BTN, BWP, BYN, BZD, CAD, CDF, CHF, CLP, CNY, COP, CRC, CUP, CVE, CZK, DJF, DKK,	PASS
OB-N04	Wrong input (should be refused)	Negative billing rates are rejected	HTTP 400	HTTP 400 msg="billingRates.0.paymentType must be one of the following values: HOURLY, FULL_DAY, HALF_DAY, FIXED, FTH, CUSTOM_HOURS, PER_DEVICE, BLENDED | billingRates.0.currency must be one of the following values: AED, AFN, ALL, AMD, ANG, AOA, ARS, AUD, AWG, AZN, BAM, BBD, BDT, BGN, BHD, BIF, BMD, BND, BOB, BRL, BSD, BTN, BWP, BYN, BZD, CAD, CDF, CHF, CLP, CNY, COP, CRC, CUP, CVE, CZK, DJF, DKK,	PASS
OB-N05	Wrong input (should be refused)	Setting ANOTHER user's billing rates (foreign userId) is refused	HTTP 403/400	HTTP 400 msg="billingRates.0.paymentType must be one of the following values: HOURLY, FULL_DAY, HALF_DAY, FIXED, FTH, CUSTOM_HOURS, PER_DEVICE, BLENDED | billingRates.0.currency must be one of the following values: AED, AFN, ALL, AMD, ANG, AOA, ARS, AUD, AWG, AZN, BAM, BBD, BDT, BGN, BHD, BIF, BMD, BND, BOB, BRL, BSD, BTN, BWP, BYN, BZD, CAD, CDF, CHF, CLP, CNY, COP, CRC, CUP, CVE, CZK, DJF, DKK,	PASS
OB-E02	Unusual situation	ABN "abc" (no format validation per checklist) Note: server accepts any ABN text — consider 11-digit validation	informational	HTTP 200 msg="SUCCESS"	PASS
OB-N06	Wrong input (should be refused)	Resume upload of a .txt is rejected	HTTP 400	HTTP 400 body={"message":"Unexpected field - resume","error":"Bad Request","statusCode":400}	PASS
OB-E03	Unusual situation	APPROVED engineer can call submitProfile again (status flips to SUBMITTED) Note: Server-side guard missing: an approved profile can be re-submitted by calling the API directly; the UI hides the route but the backend allows it.	HTTP 400/409 (not re-submitted)	HTTP 200 "SUCCESS" — status became SUBMITTED (observed 2026-10-13, account restored by admin)	FAIL
OB-U01	Normal use (should work)	PENDING engineer can open /completeYourProfile	wizard page	landing=https://dev.techdomain.com.au/jobs now=https://dev.techdomain.com.au/completeYourProfile	PASS
OB-U02	Normal use (should work)	Top line "Complete your Tech Domain profile" + "Welcome, <name>"	both	Complete your Tech Domain profile IE Welcome, Individual Engineer Here's what we need to complete your account setup. Submit Verify your Identity Tax Information Billing rates Resume Insurance Optiona	PASS
OB-U03	Unusual situation	No sidebar/topbar on the wizard (custom full page)	no "Analytics"/"Job" nav	no nav	PASS
OB-U04-VerifyyourIdentity	Normal use (should work)	Individual engineer wizard shows step "Verify your Identity"	Verify your Identity	shown	PASS
OB-U04-TaxInformation	Normal use (should work)	Individual engineer wizard shows step "Tax Information"	Tax Information	shown	PASS
OB-U04-Billingrates	Normal use (should work)	Individual engineer wizard shows step "Billing rates"	Billing rates	shown	PASS
OB-U04-Resume	Normal use (should work)	Individual engineer wizard shows step "Resume"	Resume	shown	PASS

OB-U04-Insurance	Normal use (should work)	Individual engineer wizard shows step "Insurance"	Insurance	shown	**PASS**
OB-U04-ProfileDetails	Normal use (should work)	Individual engineer wizard shows step "Profile Details"	Profile Details	shown	**PASS**
OB-U05	Normal use (should work)	"Optional" pill on Insurance / Profile Details	Optional ×2	count=4	**PASS**
OB-U06	Wrong input (should be refused)	"Submit" is disabled while required steps are incomplete	disabled	disabled	**PASS**
OB-U07	Normal use (should work)	Billing step asks for Home Address first	message shown (after opening the step)	shown	**PASS**
OB-U08	Normal use (should work)	Opening "Billing rates" shows the home-address prerequisite text	Please provide your "Home Address"	shown	**PASS**
OB-U09	Normal use (should work)	Resume step: "Resume is required…" + "Upload Resume"	both texts	shown	**PASS**
OB-U10	Normal use (should work)	Hovering an incomplete step shows "Missing details:" tooltip	Missing details	shown	**PASS**
OB-V01	Normal use (should work)	Vendor owner wizard shows the single "Company Profile" step	Company Profile; no Resume/Billing steps	Complete your Tech Domain profile VO Welcome, Vendor Owner Here's what we need to complete your account setup. Submit Company Profile Company Profile Update Reg	**PASS**
OB-V02	Wrong input (should be refused)	Vendor: Submit disabled with empty company profile	disabled	disabled	**PASS**